Panos Kolyvakis (PhD)

AI Software Engineering, Forward Deployed Engineering, Agentic AI, LLM Systems, Machine Learning

Personal Info

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Selected Projects

Q&AI: Closed-source, self-hosted **advanced RAG** app (via **Cloudflare**) that plays the exact interview moment answering your question: **LLM query augmentation**, **rank-fusion reranking** and timestamp alignment, with ~**1.3 s** searches over 284 h of video. (**Python**, **Chroma**, **LLMs**, **Whisper**)

Kyndryl Agentic AI (Framework | Policy as Code): Built the **agentic execution engine** and core of a **pure-Python agentic framework**: function calling, **MCP servers**, **MCP Registry and Gateway**, orchestration, tool discovery and execution, agent memory, concurrency; observability with **OTel**, **OpenLLMetry**, **Prometheus**, **Grafana Mimir** and dashboards; guardrails as **Rego** rules in **OPA**.

AI-To-LaTeX: A full-stack application combining **Google Programmable Search**, **web scraping**, and **LLMs** to generate and format researched content as **LaTeX** and compile it into downloadable PDFs. (**Python**, **Flask**, **JavaScript**, **HTML/CSS**, **OpenAI GPT**)

Spartan Game Engine Contributor: Open-source contributions focused on extending the engine's physics capabilities, including flotation dynamics and movement physics. (**C++**, **Physics Simulation**, **PhysX**, **Vulkan**, **GPU Computing**)

Keyboard-SnippingTool-ToGPT: A cross-platform productivity tool that captures screen content, extracts text using **OCR**, and sends it to an **LLM API** through keyboard-driven automation and a **Tkinter GUI**. (**Python**, **PowerShell**, **Bash/zsh**, **OCR**, **APIs**, **macOS Automator**)

AI leader building governed, production-grade AI for tier-1 banks and global enterprises, with a balanced foundation across classical **machine learning**, **computer vision**, **NLP**, **model training and fine-tuning**, and **agentic systems**. Currently leads agentic AI delivery at **Kyndryl**, promoted to **Associate Director** within seven months, after taking LLM and agentic systems from design to production for clients including **ABN AMRO**, **HSBC**, **SMBC** and **Deutsche Bank**.

Work Experience

Associate Director, Global AI and Data Science
Jul 2026 - present

Kyndryl, London, United Kingdom

Leading a team of engineers delivering Kyndryl's internal **agentic HR automation** platform on the **Kyndryl Agentic AI Framework**, matching contracts to staffing capacity from **enterprise data sources**. Own end-to-end delivery: **agentic and solution design**, specification, integrations, containerised deployment through **CI/CD** pipelines, and **agent reflection and evaluation**, governed by **Policy as Code** guardrails for auditable, compliant automation.

Senior Lead, Global AI and Data Science
Dec 2025 - Jul 2026

Kyndryl, London, United Kingdom

Led the delivery of **Policy as Code** for **ABN AMRO**, owning integration into the bank's systems and adapting the capability to its risk and compliance requirements. Rules extracted from business documents were converted into **Rego** and enforced through **Open Policy Agent (OPA)**, giving the bank auditable, bank-specific AI guardrails.

Senior Data Scientist AI / ML
Sept 2024 - Nov 2025

Traydstream, London, United Kingdom

Developed and productionised **ML** and **multimodal LLM** models for **document AI** in trade-finance compliance, used by clients including **SMBC**, **Staplcotn**, **HSBC** and **Deutsche Bank**. Owned fine-tuning (**full-parameter**, **LoRA**, **QLoRA**) and rigorous **LLM evaluation** for compliance-critical outputs. Built **computer-vision** (CNN object-detection) and data pipelines on **Azure VMs**, **FastAPI/Flask** APIs in **Docker**, **agentic AI** for compliance checks, and Transformer inference with **Hugging Face** and **vLLM**.

Data Scientist, R&D
Feb 2024 - Sept 2024

Twyn Holdings Limited, London, United Kingdom

Built and maintained production **LLM** and **NLP** systems on **Azure OpenAI** and **Azure Machine Learning**, including **multi-agent** and **RAG** pipelines with **LangChain**. Applied **fine-tuning** and advanced prompt engineering (**few-shot**, **chain-of-thought**) to multi-label classification.

Business Intelligence / Data Analyst
Mar 2023 - Nov 2023

Xenos Group Hotels, Zante, Greece

Applied **data analytics** to room occupancy, pricing trends and booking patterns to optimise resource allocation.

Civil Engineering Teaching Fellow
Sep 2021 - Oct 2022

School of Civil and Environmental Engineering,
The University of Edinburgh

Lectured, designed and examined two courses, **Environmental Engineering** and **Water Engineering**, and taught **Engineering Mathematics** (vector calculus, statistics) and **Engineering Design**. **Scientific programming** of wave simulations with **Python** and **finite-element methods**.

Education

PhD, Materials Science
Jul 2017 - Jul 2021

The University of Edinburgh

Thesis: Design and synthesis of porous adsorbents for water purification, with regression and computational modelling of adsorption data.

BSc Chemistry, First-class honours
Sep 2013 - Jul 2017

The University of Edinburgh

Award for **Outstanding Academic Performance**.